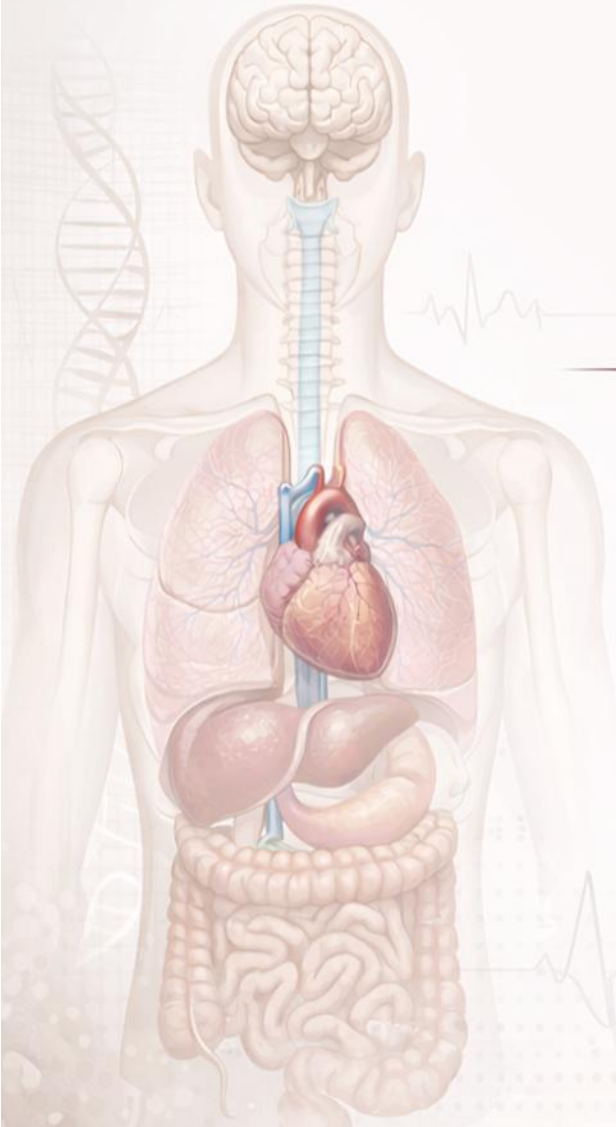


Chapter 2.
Cell Physiology

KOREA UNIVERSITY
1905

2.1 Cell Theory and Discovery

Cell Theory and Discovery

TABLE 2-1 Principles of the Cell Theory

- The cell is the smallest structural and functional unit capable of carrying out life processes.
- The functional activities of each cell depend on the specific structural properties of the cell.
- Cells are the living building blocks of all multicellular organisms.
- An organism's structure and function ultimately depend on the collective structural characteristics and functional capabilities of its cells.
- All new cells and new life arise only from preexisting cells.
- Because of this continuity of life, the cells of all organisms are fundamentally similar in structure and function.

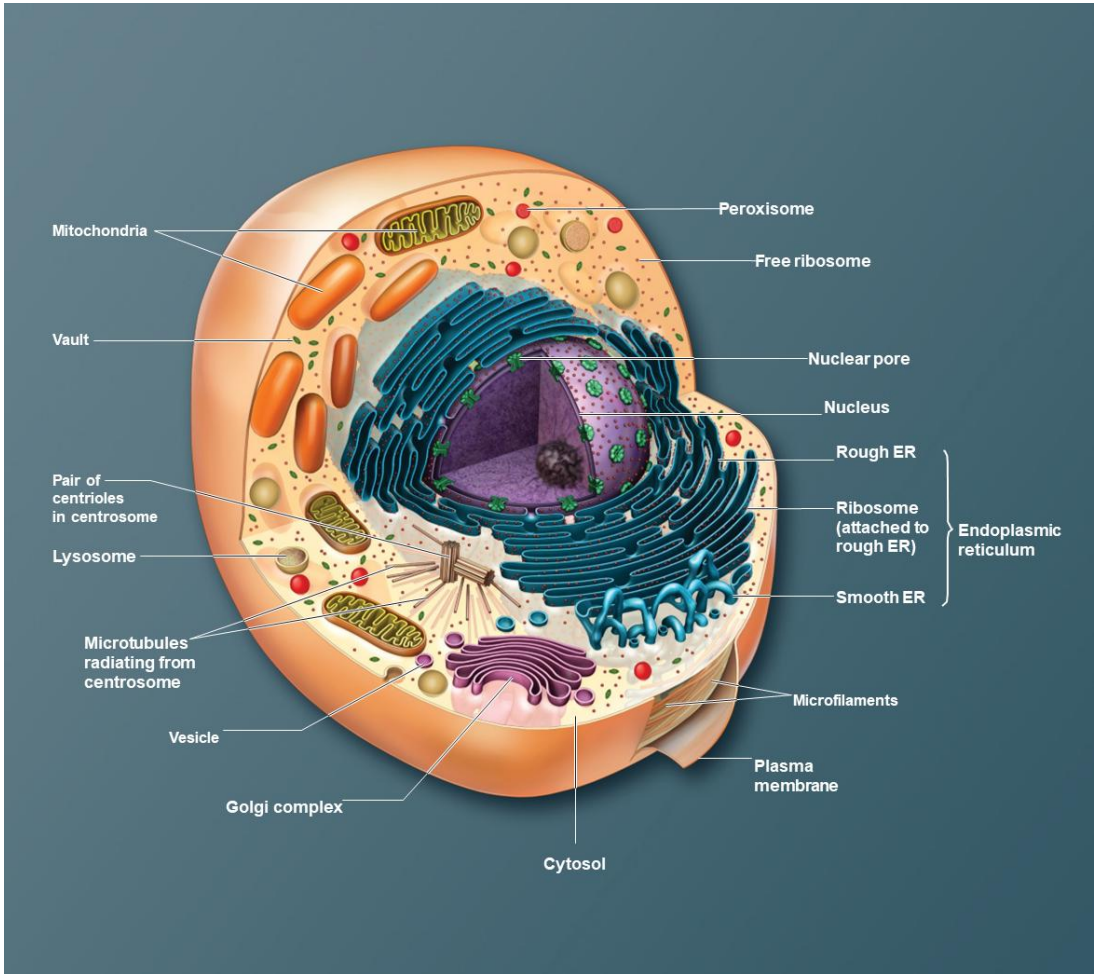
2.2 An Overview of Cell Structure

Cell Structure



https://www.youtube.com/watch?v=URUJD5NEXC8

Cell Organelles

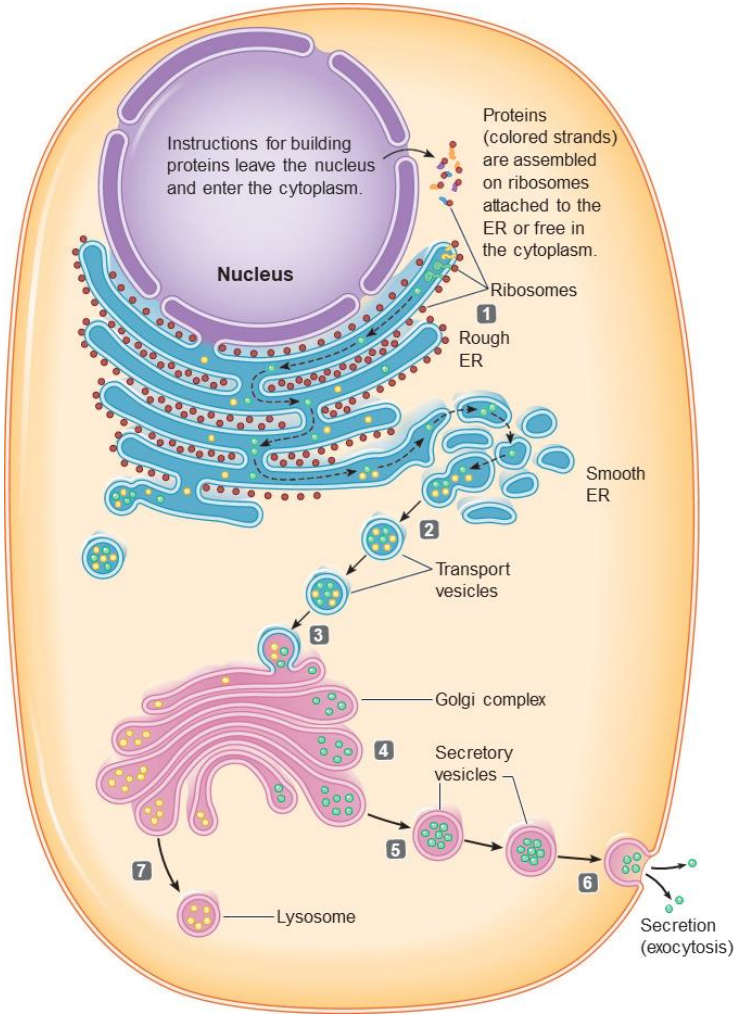


Major cell subdivisions

- Plasma membrane
- Nucleus: contains DNA
  - Important concepts: roles of RNA, human genome and proteome, epigenetics, and lipidome
- Cytoplasm: consists of various organelles, the cytoskeleton, and the cytosol
  - Cytoplasm is the portion of the cell interior not occupied by the nucleus
  - Cytosol is specifically the water-based, jelly-like fluid within the cytoplasm that surrounds these organelles
  - Cytosol vs cytoplasm

2.3 Endoplasmic Reticulum and Segregated Synthesis

Endoplasmic Reticulum and Segregated Synthesis



1 The rough ER synthesizes proteins to be secreted to the exterior or to be incorporated into plasma membrane or other cell components.

2 The smooth ER packages the secretory product into transport vesicles, which bud off and move to the Golgi complex.

3 The transport vesicles fuse with the Golgi complex, open up, and empty their contents into the closest Golgi sac.

4 The newly synthesized proteins from the ER travel by vesicular transport through the layers of the Golgi complex, which modifies the raw proteins into final form and sorts and directs the finished products to their final destination by varying their wrappers.

5 Secretory vesicles containing the finished protein products bud off the Golgi complex and remain in the cytosol, storing the products until signaled to empty.

6 On appropriate stimulation, the secretory vesicles fuse with the plasma membrane, open, and empty their contents to the cell's exterior. Secretion has occurred by exocytosis, with the secretory products never having come into contact with the cytosol.

7 Lysosomes also bud from the Golgi complex.

Segregated protein synthesis: spatial separation of protein production in cells

- Cytosolic ribosomes produce intracellular proteins
- ER-bound ribosomes produce secreted and membrane proteins

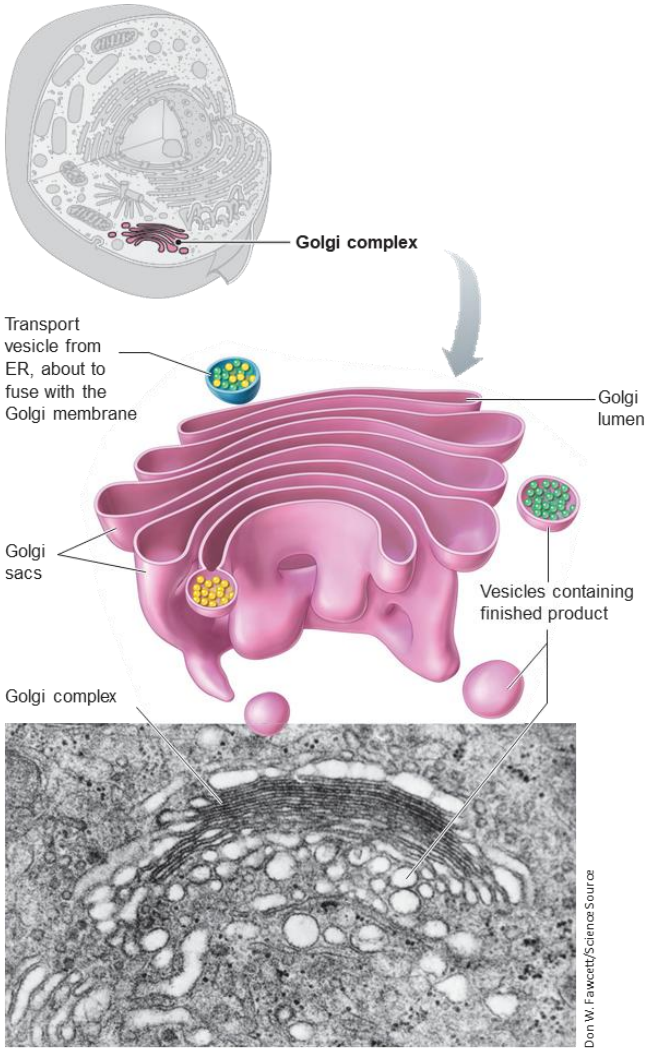
Endoplasmic Reticulum and Segregated Synthesis

- Endoplasmic reticulum (ER) is an elaborate fluid-filled membranous system
  - Distributed extensively throughout the cytosol
  - Primary function: produce proteins and lipids
  - Rough ER synthesizes proteins for secretion and membrane construction
  - Smooth ER packages new proteins in transport vesicles
  - Misfolded proteins are destroyed by the ubiquitin–proteasome pathway
  - Ubiquitin system labels flawed proteins for destruction, and other damaged or unneeded intracellular proteins for degradation

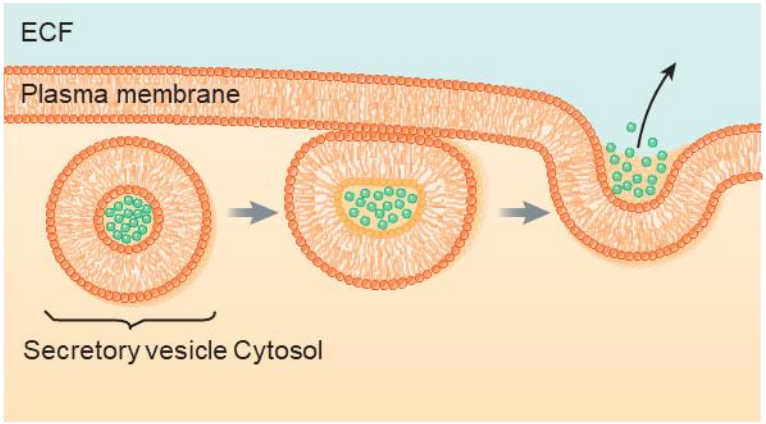
2.4 Golgi Complex and Exocytosis

Golgi Complex and Exocytosis

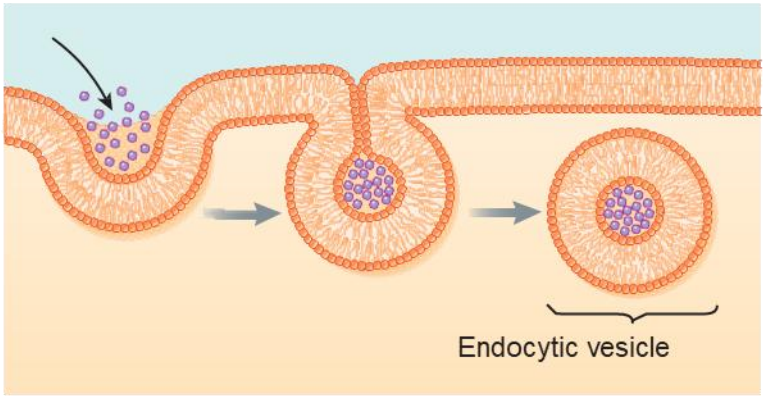
- Golgi complex consists of a stack of flattened, slightly curved, membrane-enclosed sacs
  - Unlike ER, they do not come into contact with one another
  - Vesicular transport from one Golgi sac to the next: accomplished through action of membrane-curving coat protein I (COPI)
  - In secretory cells, the Golgi complex packages proteins for export by exocytosis



Exocytosis and Endocytosis



(a) Exocytosis: A secretory vesicle fuses with the plasma membrane, releasing the vesicle contents to the cell exterior. The vesicle membrane becomes part of the plasma membrane.

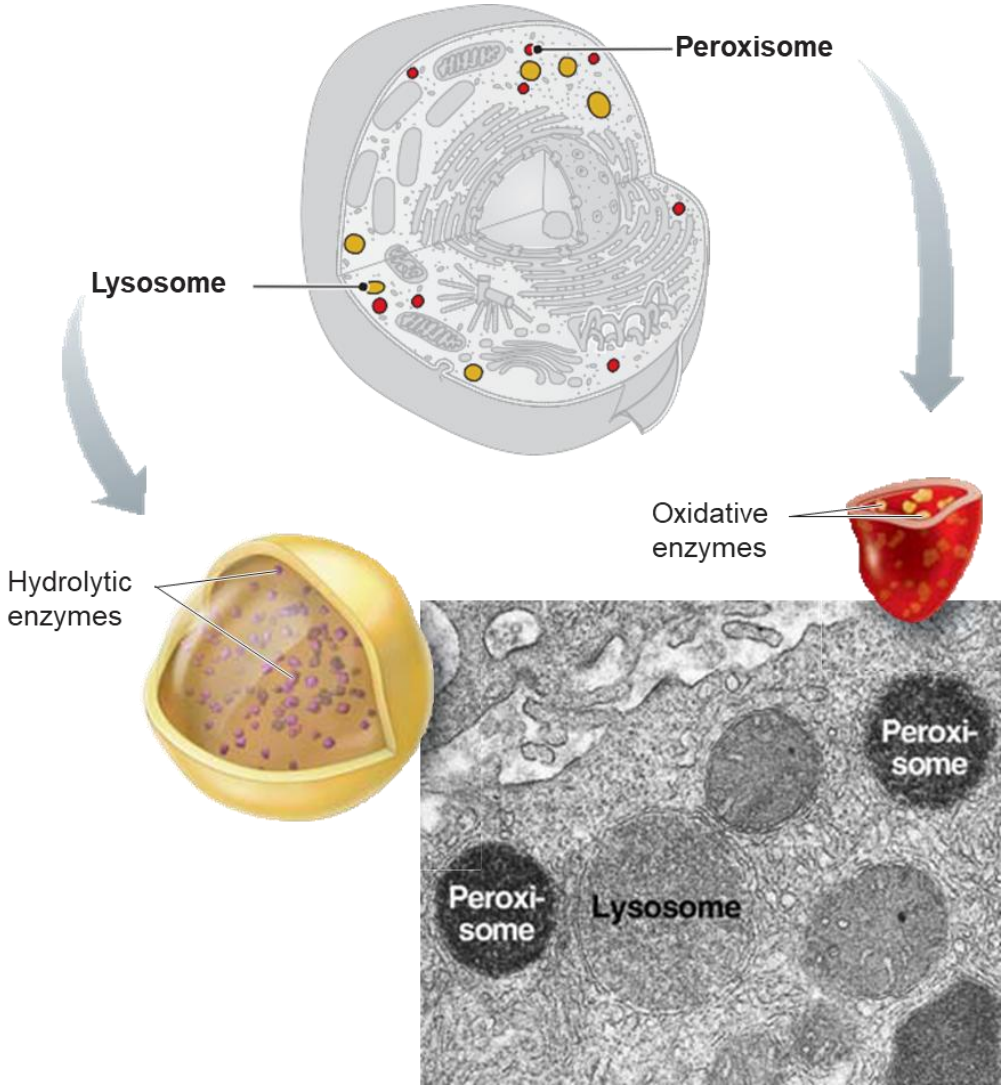


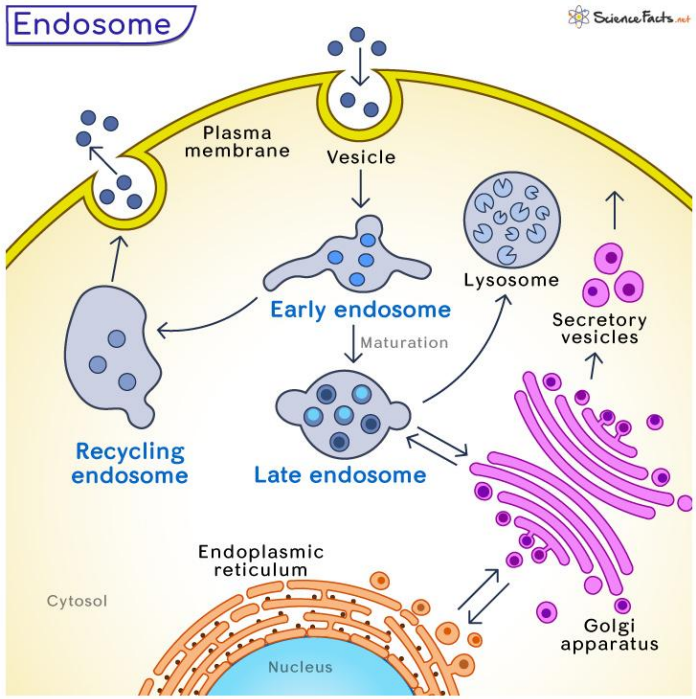
(b) Endocytosis: Materials from the cell exterior are enclosed in a segment of the plasma membrane that pockets inward and pinches off as an endocytic vesicle.

- Secretion releases the product made by the cell to the external environment (ECF)
  - Specialized secretory cells: endocrine cells (hormones), digestive gland cells (digestive enzymes)
  - Secretory vesicles bud off from Golgi stacks
  - Exocytosis releases cell-produced substances to the ECF (primary mechanism for secretion)
  - Secretory vesicles only bind to plasma membrane, not to other organelle membranes
- Endocytosis refers to the internalization of extracellular material into the cell
  - Pinocytosis
  - Receptor-mediated endocytosis
  - Phagocytosis

2.5 Lysosomes and Endocytosis

Lysosomes and Peroxisomes





Lysosomes and Endocytosis

- Lysosomes: small, membrane-enclosed, degradative organelles
  - Digest extracellular material brought into the cell by phagocytosis and remove worn-out organelles
  - Endocytosis: internalization of extracellular material within a cell
    - Pinocytosis: nonselective uptake of ECF
    - Receptor-mediated endocytosis: selective uptake of specific large molecules
    - Phagocytosis: internalization of large particles
  - Endosome: membrane-bound vesicles that act as central sorting stations for endocytosed materials

2.6 Peroxisomes and Detoxification

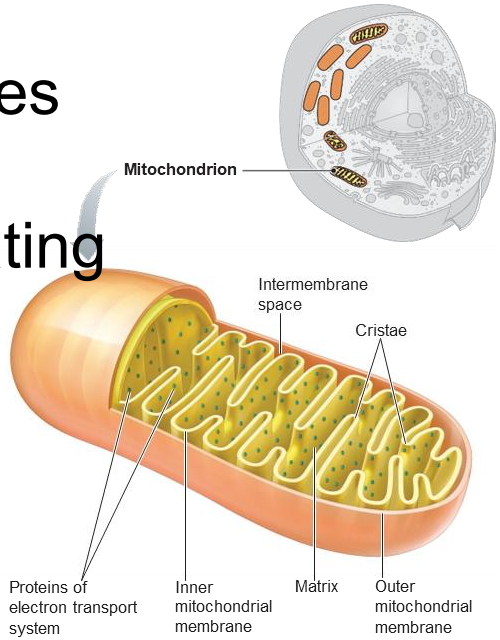
Peroxisomes and Detoxification

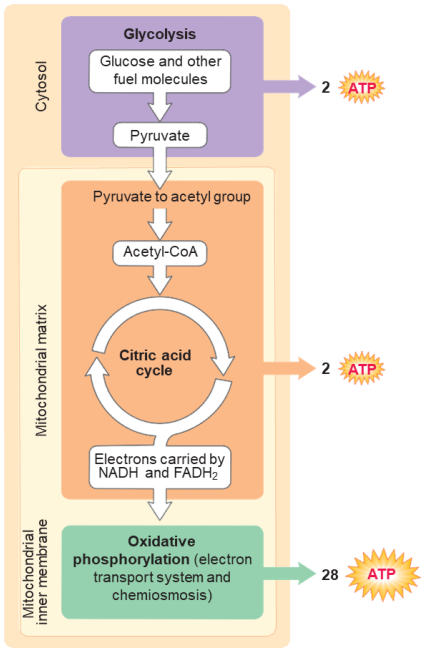
- Peroxisomes are membranous organelles which produce and degrade hydrogen peroxide (H_2O_2) while degrading potentially toxic molecules
  - House oxidative enzymes that detoxify various wastes
  - Oxidative enzymes (oxidases) use oxygen (O_2) to strip hydrogen from certain organic molecules and produce hydrogen peroxide
  - Hydrogen peroxide is degraded by catalase

2.7 Mitochondria and ATP Production

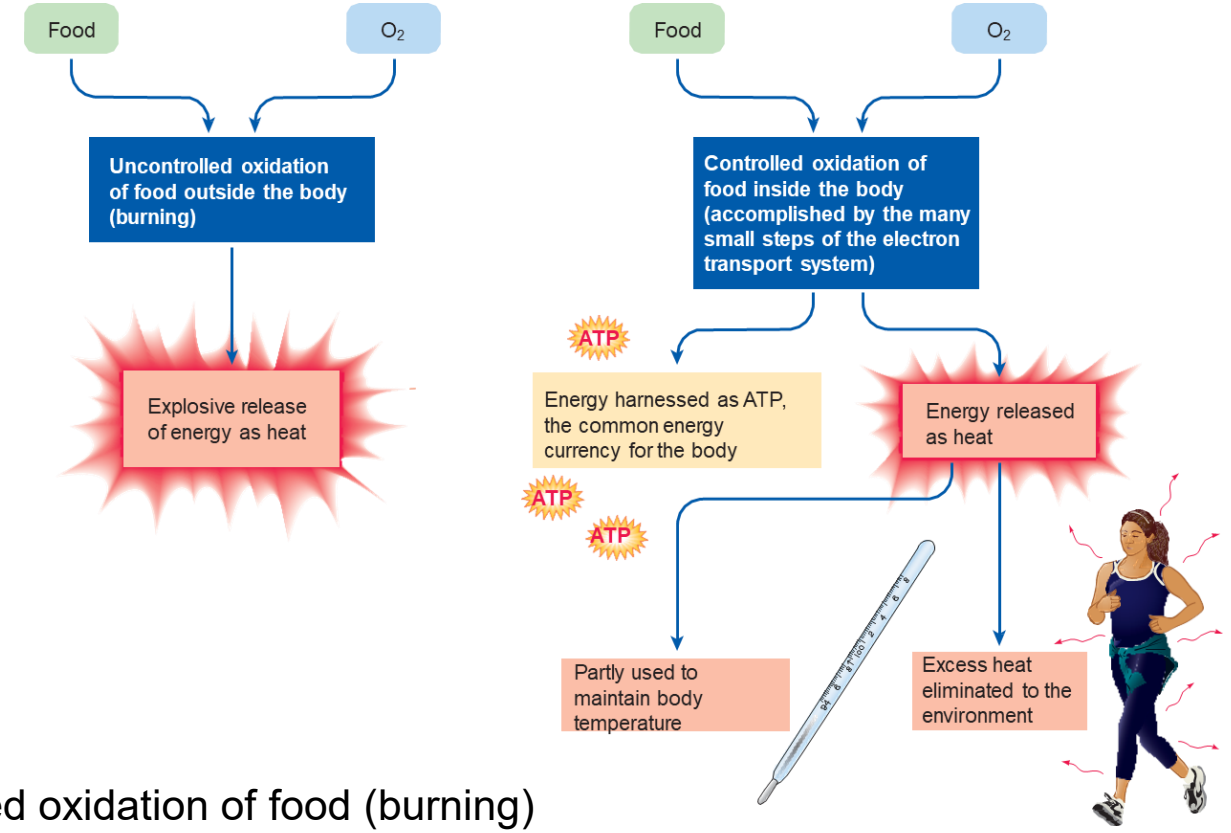
Mitochondria and ATP Production

- Mitochondria: energy organelles of the cell
  - Extract energy from food nutrients and transform it into a usable form for cellular activities
  - Enclosed by two membranes that form the cristae
  - Play a major role in generating ATP (cellular respiration)
    - Via processes including glycolysis (cytosol), pyruvate oxidation, citric acid cycle, oxidative phosphorylation (mitochondria)





Uncontrolled vs controlled oxidation of food

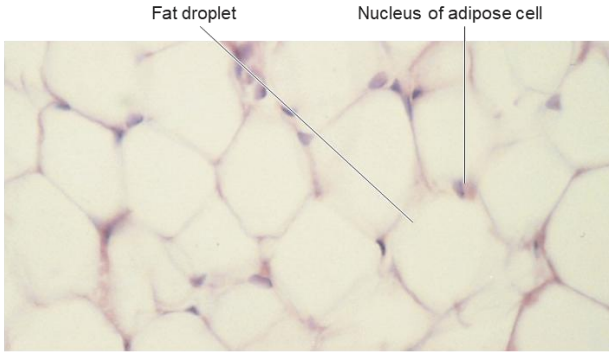


- Uncontrolled oxidation of food (burning)
  - The energy is released rapidly and explosively as heat, no usable energy is captured
- Controlled oxidation of food in the body
  - The oxidation reaction occurs inside cells mainly through electron transport chain in mitochondria
  - ATP production → used for cellular work (movement, biosynthesis, transport)
  - Heat generation → helps maintain body temperature
  - Excess heat → released to the environment

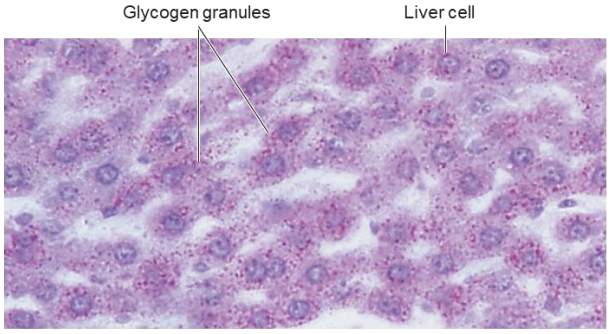
2.9 Cytosol: Cell Gel

Cytosol: Cell Gel

- Cytosol: semiliquid portion of cytoplasm surrounding organelles
  - Categories of activities associated with cytosol
    - Enzymatic regulation of intermediary metabolism
    - Ribosomal protein synthesis
    - Storage of fat, carbohydrate, and secretory vesicles (inclusions)
    - Inclusions: nonpermanent mass of stored materials, not surrounded by membrane. Its presence depends on cell types and circumstances



(a) Fat storage in adipose cells



(b) Glycogen storage in liver cells

2.10 Cytoskeleton: Cell “Bone and Muscle”

Cytoskeleton

- Cytoskeleton: complex protein network that supports and organized the cell organelles and control their movement
  - Elements
    - Microtubules: help maintain asymmetric cell shapes and play a role in complex cell movements
    - Microfilaments: important to cellular contractile systems and as mechanical stiffeners
    - Intermediate filaments: important in cell regions subject to mechanical stress
  - Links parts of the cell
    - Responsible for the particular shape, rigidity, and spatial geometry of each different cell type
    - mechanical communications system
    - directs intracellular transport and regulates numerous cellular movements